

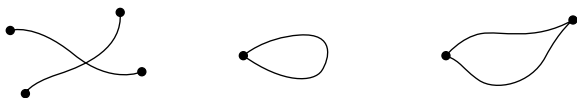
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Graphs

The theory of graphs, otherwise known as networks, is a branch of mathematics which finds much application both within mathematics and in science. There are many books where the reader can find evidence to support this claim, for example those by Berge, Gross & Yellen, Diestel, Wilson & Beinecke and Bollobás listed in the References. Our motivation for touching on the subject here is different. Many of the ideas which we shall encounter later can be met, in a diluted form, in the simpler situation of graph theory, and that is the reason why we study graphs first. The problems studied by graph theorists are usually specifically applicable to graphs, and do not have sensible analogues in the more general theory of ‘simplicial complexes’ which we shall study. It is hardly surprising, therefore, that the theory we shall develop has little to say of interest to graph theorists, and it is partly for this reason that an occasional excursion is made, in this chapter and elsewhere, into genuine graph-theoretic territory. This may give the reader some idea of the difficult and interesting problems which lie there, but which the main concern of this book, namely homology theory, can scarcely touch.

Abstract graphs and realizations

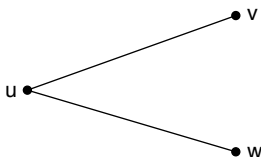
A graph is intuitively a finite set of points in space, called the vertices of the graph, some pairs of vertices being joined by arcs, called the edges of the graph. Two arcs are assumed to meet, if at all, in a vertex, and it is also assumed that no edge joins a vertex to itself and that two vertices are never connected by more than one edge. (Such a graph is often called a simple graph.) That is, we do not allow



The positions of the vertices and the lengths of the edges do not concern us; what is important is the number of vertices and the pairs of vertices which are connected by an edge. Thus the only information we need about an edge is the names of the vertices which it connects. This much information can be summed up in the following ‘abstract’ definition; we shall return to ‘reality’ shortly.

Definition 1.1. An *abstract graph* is a pair (V, E) where V is a finite set and E is a set of unordered pairs of distinct elements of V . Thus an element of E is of the form $\{v, w\}$ where v and w belong to V and $v \neq w$. The elements of V are called *vertices* and the element $\{v, w\}$ of E is called the *edge* joining v and w (or w and v).

Note that E may be empty: it is possible for no pair of vertices to be joined by an edge. Indeed V may be empty, in which case E certainly is – this is the empty graph with no vertices and no edges. In any case if V has n elements then E has at most $\binom{n}{2} = \frac{1}{2}n(n-1)$ elements. An abstract graph with the maximum number of edges is called *complete*: every pair of vertices is joined by an edge.



As an example of an abstract graph let $V = \{u, v, w\}$ and $E = \{\{u, v\}, \{u, w\}\}$. This abstract graph (V, E) can be pictured, or ‘realized’ as we shall say in a moment, by the diagram.

In many circumstances it is convenient to speak of graphs in which each edge has a direction or orientation (intuitively an arrow) attached to it. For example in an electrical circuit carrying direct current each edge (wire) has a direction, namely the direction in which current flows.

Definition 1.2. An *abstract oriented graph* is a pair (V, E) where V is a finite set and E is a set of ordered pairs of distinct elements of V with the property that, if $(v, w) \in E$ then $(w, v) \notin E$. The elements of V are called *vertices* and the element (v, w) of E is called the *edge* from v to w .

Associated with any abstract oriented graph there is an abstract graph, obtained by changing ordered to unordered pairs. The two graphs have the

same number of vertices, and the same number of edges on account of the assumption that (v, w) and (w, v) never both belong to E .

We now return to graphs ‘realized’ in space.

Definition 1.3. Let (V, E) be an abstract graph. A *realization* of (V, E) is a set of points in a real vector space $\mathbb{R}^N$, one point for each vertex, together with straight segments joining precisely those pairs of points which correspond to edges. The points are called *vertices* and the segments *edges*; the realization is called a *graph*. We require two ‘intersection conditions’ to hold: (i) two edges meet, if at all, in a common end-point, and (ii) no vertex lies on an edge except at one of its end-points. The edge joining v and w will be denoted (vw) , or (wv) .

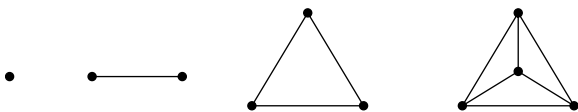
Note that the edges of a graph are straight; that this is no restriction for graphs in $\mathbb{R}^3$ follows from Theorem 1.4 below. We shall turn to the question of whether it is a restriction for graphs in $\mathbb{R}^2$ (planar graphs) in 1.34.

A realization of an abstract oriented graph, called an *oriented graph*, is just a realization of the associated abstract graph, together with the ordering on each edge of the graph. In practice this ordering is indicated by an arrow from the first to the second vertex of the edge:



is the edge from v to w , which will be denoted (vw) .

The complete abstract graphs with up to four vertices have realizations in $\mathbb{R}^2$:



The reader can probably convince himself that the complete abstract graph with five vertices cannot be realized in $\mathbb{R}^2$ (see Proposition 1.32); however it can be realized in $\mathbb{R}^3$, as indeed can every abstract graph.

Theorem 1.4. Every abstract graph can be realized in $\mathbb{R}^3$.

Proof Let C be the subset of $\mathbb{R}^3$ defined by

$$C = \{(x, x^2, x^3) : x \in \mathbb{R}\}.$$

(C is called a ‘twisted cubic’.) I claim that no four points of C are coplanar (it follows from this that no three points of C are collinear). To prove the claim

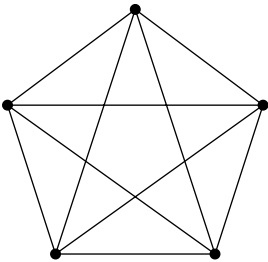
take four points P_1, P_2, P_3, P_4 of C , given respectively by distinct real numbers x_1, x_2, x_3, x_4 . They are coplanar if and only if the vectors $P_1 - P_2, P_1 - P_3, P_1 - P_4$ are linearly dependent,

i.e.
$$\begin{vmatrix} x_1 - x_2 & x_1^2 - x_2^2 & x_1^3 - x_2^3 \\ x_1 - x_3 & x_1^2 - x_3^2 & x_1^3 - x_3^3 \\ x_1 - x_4 & x_1^2 - x_4^2 & x_1^3 - x_4^3 \end{vmatrix} = 0.$$

However this determinant vanishes if and only if the x 's are not distinct, and this proves the claim.

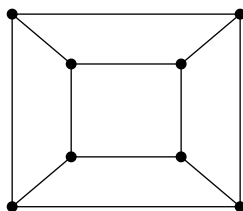
To realize an abstract graph (V, E) in $\mathbb{R}^3$ take n distinct points $P_1, \dots, P_n$ of C , one for each vertex, and join precisely those pairs of points which correspond to elements of E by straight segments. It remains to verify that the intersection conditions hold. Let P_iP_j and P_kP_ℓ be two (distinct) segments; certainly $i \neq j$ and $k \neq \ell$ and we can assume $j \neq \ell$. If $i = k$, so that the segments have a common end-point, it is still true that P_i, P_j and P_ℓ are distinct and so not collinear; hence the segments do not have any more points in common. If on the other hand $i \neq k$ then all four P 's are distinct and so not coplanar; hence the segments do not meet at all. This shows that (i) of 1.3 holds; (ii) follows immediately from the fact that no three points of C are collinear. □

Remarks 1.5. (1) When drawing graphs it is often convenient to draw a figure in the plane with heavy dots for vertices and straight lines (or even curved ones) for edges, possibly intersecting at points other than end-points. For example the complete graph with five vertices might be drawn as in the diagram below. This is to be regarded as a recipe for constructing an abstract graph which Theorem 1.4 assures us can be realized in $\mathbb{R}^3$. When the edges are straight, as in the diagram, it can also be regarded as the projection into the plane of a realization in $\mathbb{R}^3$.



(2) The proof of 1.4 still works if the definition of ‘abstract graph’ is relaxed to the extent of allowing the set V of vertices to be countable.

Of course the same abstract graph can be realized in many ways. The graph in $\mathbb{R}^3$ formed by the edges and vertices of a cube, and the planar graph (see the diagram below) are realizations of the same abstract graph. This, and the fact that the precise names given to the vertices of an abstract graph are unimportant, suggests the following definition.

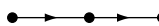
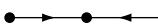


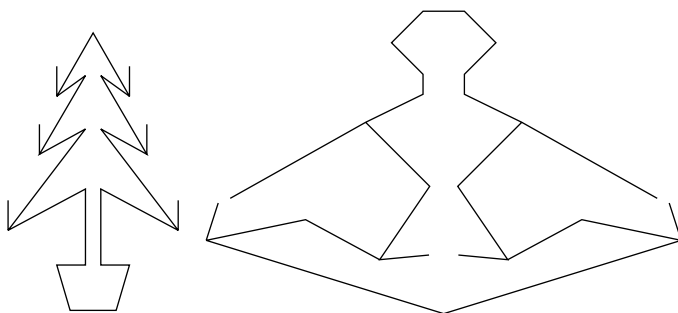
Definition 1.6. Two abstract graphs (V, E) and (V', E') are called *isomorphic* if there is a bijective map $f : V \rightarrow V'$ such that

$$\{v, w\} \in E \Leftrightarrow \{f(v), f(w)\} \in E'.$$

(Thus f is simply a relabelling of vertices which preserves edges.) Two graphs are called *isomorphic* if they are realizations of isomorphic abstract graphs.

The corresponding definitions for oriented graphs are obtained by replacing unordered pairs $\{ , \}$ by ordered pairs $(,)$.

Thus  and  are isomorphic graphs but  and  are not isomorphic oriented graphs. The Christmas tree and Cossack dancer are isomorphic.

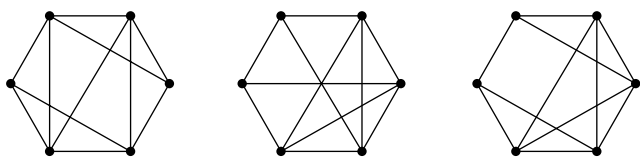


(For once, vertices are not emphasized with dots.)

To prove two graphs are isomorphic we have to construct an explicit map from the vertex set of one to the vertex set of the other which is bijective and preserves edges. To prove two graphs are not isomorphic we look at properties

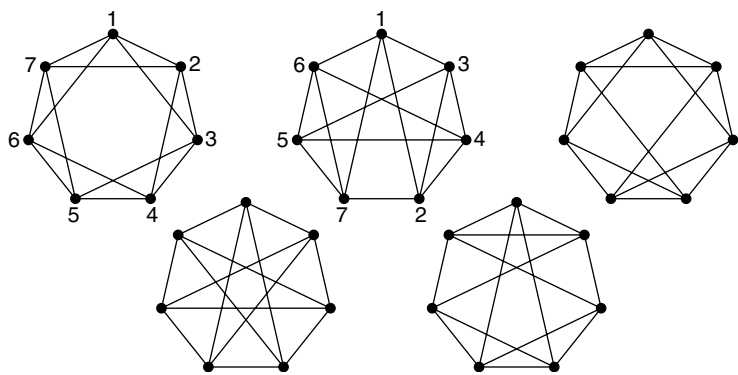
of graphs which are invariant under isomorphisms and try to find such a property possessed by one graph but not by the other. The simplest such property is the number of vertices or edges; next simplest is the orders of the vertices. The *order* of a vertex is the number of edges with the vertex as an end-point, and clearly two isomorphic graphs must have the same number of vertices of any given order. Slightly more subtle methods are needed for the following two examples.

Examples 1.7. (1X) Exactly two of the following three graphs are isomorphic. Which two? (Compare remark 1.5(1).)



[Hint. Look at the vertices of order 3.]

(2X) The first two graphs below are isomorphic, and the vertices are labelled to show this. Which other pairs are isomorphic? (Again see 1.5(1).)



[Hint. Count triangular paths.]

★ Kirchhoff's laws

The discussion which follows is intended to show how one of the fundamental properties of a graph, its 'cyclomatic number', arises naturally in the study of direct current (D.C.) electrical circuits. The development of graph theory proper continues at 1.8.

We can associate a graph with a D.C. circuit, replacing terminals by vertices and wires by edges. An edge joining a vertex to itself or two or more edges connecting the same pair of vertices can easily be eliminated by the addition of extra vertices which do not essentially change the graph from an electrical point of view. We can also assume that the graph is all in one piece, since separate D.C. circuits can be treated separately. Each edge has associated with it positive real number called the resistance, and some edges will in addition contain sources of power. The edges are oriented (arbitrarily) and the object is to calculate the current in each edge; a negative answer indicates that the current is flowing against the chosen orientation.

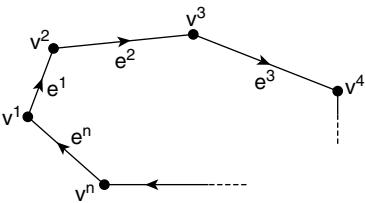
Kirchhoff's laws provide linear equations for the currents. These equations fall into two types:

(K1) There is one equation for each closed loop[†] in the graph. A loop is first given a direction, one way round or the other, and in the resulting equation the only currents which appear are those along edges in the loop. The coefficient of a current is the resistance of the edge if the orientation of the edge agrees with that of the loop; otherwise it is minus the resistance. The 'right-hand side' of the equation is the total electro-motive force (e.m.f.) in the loop, with an appropriate convention on signs.

(K2) There is one equation for each vertex, expressing the fact that the total current flowing into the vertex is zero (a current flowing out being counted negatively).

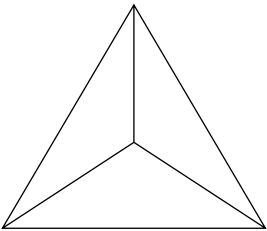
Suppose that there are α_0 vertices and α_1 edges; then the α_0 equations (K2) are not independent since if we add them all up the sum vanishes identically. However it is not hard to verify that any $\alpha_0 - 1$ of them are independent. Since there are α_1 unknowns it follows that in order to determine the currents we shall need at least $\alpha_1 - \alpha_0 + 1$ independent equations (K1) – possibly more, since there may be relations between these and the (K2) equations. In fact, there are exactly $\alpha_1 - \alpha_0 + 1$ independent equations (K1) (that is, $\alpha_1 - \alpha_0 + 1$ 'independent loops') and these are also independent of (K2) and so (K1) and (K2) between them uniquely determine the currents. A proof of these facts is sketched in an appendix at the end of the chapter, where we have more notation at our disposal. In the meantime here is a simple proof for the case of a graph with a single loop. For convenience assume the edges $e^1, \dots, e^n$ of the loop are all oriented in the same direction round the loop. Let $v^1, \dots, v^n$ be the vertices round the loop; there may be other vertices and edges not in the loop but these do not affect the argument.

[†] Here we use 'loop' to mean a simple closed path round the edges; a precise definition is in 1.8 below. Beware that some authors use 'loop' for an edge joining a vertex to itself, something we forbid here.



Suppose that the (K1) equation of the loop is a linear combination of (K2) equations, the coefficient of the (K2) equation for v^i being $\lambda_i, i = 1, \dots, n$. The current along edge e^1 can occur only in the (K2) equations for v^1 and v^2 , so that the coefficient of this current in the (K1) equation of the loop must be $\lambda_2 - \lambda_1$. Since this is positive (resistances are positive and e^1 is oriented round the loop) we have $\lambda_2 > \lambda_1$. In a similar way $\lambda_3 > \lambda_2, \lambda_4 > \lambda_3, \dots, \lambda_n > \lambda_{n-1}, \lambda_1 > \lambda_n$. Hence $\lambda_1 > \lambda_1$, a contradiction.

The ‘number of independent loops’ $\alpha_1 - \alpha_0 + 1$ arising from Kirchhoff’s analysis of electrical circuits was called the ‘cyclomatic number’ of the circuit by James Clerk Maxwell.



As a simple example, the graph above has cyclomatic number 3 and it is certainly very plausible that every loop is a ‘combination’ of the three loops which bound the regions inside the big triangle. ★

Maximal trees and the cyclomatic number

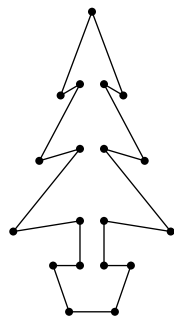
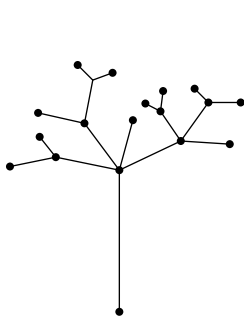
Definitions 1.8. A *path* on a graph G from v^1 to v^{n+1} is a sequence of vertices and edges

$$v^1 e^1 v^2 e^2 \dots v^n e^n v^{n+1}$$

where $e^1 = (v^1 v^2), e^2 = (v^2 v^3), \dots, e^n = (v^n v^{n+1})$. (See 1.3 for notation. The path goes from v^1 to v^{n+1} along the edges $e^1, \dots, e^n$.) Here $n \geq 0$: for $n = 0$ we

just have v^1 and no edges. If G is oriented, we only require that $e^i = (v^i v^{i+1})$ or $(v^{i+1} v^i)$ for $i = 1, \dots, n$; that is, the edges along the path do not have to point in the ‘right’ direction. The path is called *simple* if the edges $e^1, \dots, e^n$ are all distinct, and the vertices $v^1, \dots, v^{n+1}$ are all distinct except that possibly $v^1 = v^{n+1}$. If the simple path has $v^1 = v^{n+1}$ and $n > 0$ it is called a *loop*; thus in fact a loop always has $n \geq 3$. We shall regard two loops as equal when they consist of the same vertices and the same edges.

A graph G is called *connected* if, given any two vertices v and w of G there is a path on G from v to w . For any non-empty graph G , a *component* of G consists of all the edges and vertices which occur in paths starting at some particular vertex of G . Thus a non-empty connected graph has exactly one component. A graph which is connected and has no loops is called a *tree*.

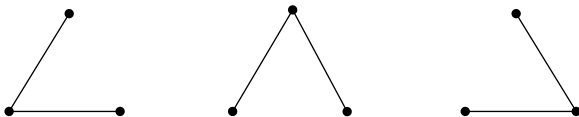


Examples 1.9. (1)

is a tree, but

is not.

(2) Given n points in space – say n points on the twisted cubic of the proof of 1.4 – we can ask how many trees there are with those points as vertices. For instance when $n = 3$, the answer is 3:



However these are all isomorphic, and a more interesting question is: how many trees are there no two of which are isomorphic? The former question has the answer n^{n-2} ($n \geq 1$), but the general answer to the latter question is unknown. (The first result is known as Cayley’s Theorem, and it is discussed in books on graph theory such as those listed at the beginning of this chapter.)

(3) If there is a path on G from v to w then by elimination of any repetitions a simple path can be constructed on G from v to w .

(4X) An edge e of a graph is part of some loop on the graph if and only if the removal of e does not increase the number of components of the graph.

Definitions 1.10. Given a graph G , a graph H is called a *subgraph* of G if the vertices of H are vertices of G and the edges of H are edges of G . Also H is called a *proper subgraph* of G if in addition $H \neq G$. Any graph G will have a subgraph which is a tree (e.g. the empty subgraph is a tree!) so that the set $\mathcal{T}$ of subgraphs of G which are trees will have maximal elements. That is, there will be at least one $T \in \mathcal{T}$ such that T is not a proper subgraph of any $T' \in \mathcal{T}$. Such a T is called a *maximal tree* for G . (Some authors say *spanning tree*.)

Maximal trees are mainly of interest for connected graphs, and we have the following result.

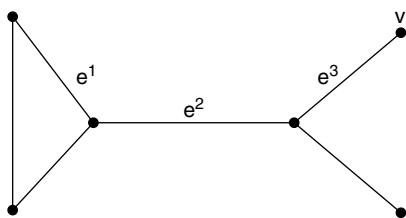
Proposition 1.11. *Let G be a connected graph. A subgraph T of G is a maximal tree for G if and only if T is a tree containing all the vertices of G .*

Proof We may suppose G is non-empty. Suppose that T is a maximal tree for G . Certainly T is then a non-empty tree; suppose that a vertex v of G fails to belong to T . Choose a vertex w of T and a path from v to w on G . Let v^{i+1} be the first vertex of this path which belongs to T ; then the previous vertex v^i and the edge $e^i = (v^i v^{i+1})$ will not belong to T . Adding v^i and e^i to T we obtain say T' , a larger subgraph of G than T . Furthermore, e^i cannot be part of any loop on T' since one of its end-points, v^i , does not belong to any edge of T' other than e^i . Thus T' is a tree and T is not maximal.

Suppose conversely that T is a subgraph of G which is a tree and contains all the vertices of G . Suppose that H is a subgraph of G with T a proper subgraph of H . Since T already has all the vertices of G there must be an edge, $e = (vw)$ say, in H but not in T . Since T is connected there is a simple path on T from v to w (compare 1.9(3)); adding e to the end of this path gives a loop in H . Hence H is not a tree and T is maximal. $\square$

Examples 1.12. (IX) Given a connected graph G and a subgraph T which is a tree, T is a subgraph of a maximal tree for G . Indeed, given a subgraph H of G which contains no loops (but is not necessarily connected), H is a subgraph of a maximal tree for G . (Use the method of the first paragraph of the proof of 1.11.)

(2X) A vertex v of an edge (vw) in a graph G is called *free* (or *terminal*) if v is not a vertex of any edge of G other than e . An *inner edge* of a graph is an edge neither of whose end-points is a free vertex.



Thus in the diagram e^1 and e^2 are inner edges, e^3 is not and v is a free vertex of e^3 . A graph with at least one edge, all of whose edges are inner, is bound to have a loop (since the number of edges is finite); it follows that a tree with at least one edge always possesses a non-inner edge. Removing a non-inner edge e of a graph, together with a free vertex of e , is called an *elementary collapse*, and a sequence of these is called a *collapse*. (We shall meet this idea again in 3.30.) Since a collapse takes a tree to a smaller tree it follows that any tree collapses to a single vertex. Indeed, the converse is true: if a graph collapses to a single vertex, then it must have been a tree.

The following definition and theorem connect the idea of maximal tree with the number $\alpha_1 - \alpha_0 + 1$ introduced during the discussion of Kirchhoff's laws.

Definition 1.13. Let G be a connected graph with $\alpha_0 = \alpha_0(G)$ vertices and $\alpha_1 = \alpha_1(G)$ edges. The *cyclomatic number* of G is the integer

$$\mu = \mu(G) = \alpha_1 - \alpha_0 + 1.$$

Theorem 1.14. Let G be a graph with cyclomatic number μ . Then

- (1) $\mu \geq 0$.
- (2) $\mu = 0$ if and only if G is a tree.
- (3) Every maximal tree for G has $\alpha_1 - \mu$ edges. That is, every maximal tree for G can be obtained from G by removing μ suitable edges (and no vertices). These edges must be inner edges since T is connected.
- (4) Removing fewer than μ edges from G cannot produce a tree; removing more than μ edges always disconnects G .

Proof The proof of (1) and (2) is by induction on α_1 . Clearly (1) and (2) are true when $\alpha_1 = 0$; suppose they are true when $\alpha_1 < k$, where $k \geq 1$, and let G have k edges.

(i) Suppose G is a tree. Then G has a non-inner edge (compare 1.12(2)), e say. Removing e and a free vertex of e from G leaves a tree G' with $k - 1$ edges. By the induction hypothesis, $\mu(G') = 0$, and by construction $\mu(G') = \mu(G)$. Hence $\mu(G) = 0$.

(ii) Suppose G is not a tree. Then G has an inner edge, namely any edge of a loop of G ; let G' be obtained by removing this edge (and no vertices). Removing an edge from a loop cannot disconnect G , so that G' is connected and by the induction hypothesis $\mu(G') \geq 0$. By construction $\mu(G') = \mu(G) - 1$; hence $\mu(G) \geq 1$.

This completes the proof by induction of (1) and (2).

To prove (3), let T be a maximal tree for G , so that by 1.11 $\alpha_0(G) = \alpha_0(T)$. Using this and $\mu(T) = 0$ it follows easily that $\alpha_1(T) = \alpha_1(G) - \mu(G)$.

To prove (4), note that removing p edges (and no vertices) from G leaves either a disconnected graph or else a connected graph with cyclomatic number $\mu(G) - p$. From this and (1) and (2) we can deduce (4). $\square$

Remarks 1.15. (1) Removing μ edges at random may not leave a connected graph, hence certainly not a tree. However if it does leave a connected graph then this must be a maximal tree: it will contain all the vertices and have cyclomatic number $\mu - \mu = 0$.

(2) Let G be a connected graph with $\mu(G) = 1$. Then G has exactly one loop. (Proof: G has at least one loop since $\mu(G) > 0$. If there are two loops let e be an edge of one loop not in the other. Removing e from G leaves G connected, and still containing a loop, but it also lowers μ by one to zero. This is a contradiction.)

1.16. The basic loops

Let G be a connected graph with cyclomatic number μ and let $e^1, \dots, e^\mu$ be edges whose removal from G leaves a maximal tree T (see 1.14 (3)). We shall denote by $T + e^i$ the graph obtained from T by replacing e^i . Then $\mu(T + e^i) = 1$, and by 1.15(2) there is a unique loop ℓ^i on $T + e^i$. We shall prove that in some sense these loops ‘generate’ all loops on G ; one consequence of this is that the Kirchhoff equations for $\ell^1, \dots, \ell^\mu$ are enough to determine all equations of type (K1). Before doing this we shall generalize the idea of loop.

Chains and cycles on an oriented graph

Let G be an oriented graph. Given a path

$$v^1 e^1 v^2 e^2 \dots v^n e^n v^{n+1}$$

on G we can associate with it a formal sum

$$\varepsilon_1 e^1 + \varepsilon_2 e^2 + \dots + \varepsilon_n e^n$$

where $\varepsilon_i = +1$ if $e^i = (v^i v^{i+1})$ and $\varepsilon_i = -1$ if $e^i = (v^{i+1} v^i)$. Thus the formal sum is obtained by travelling along the path from v^1 to v^{n+1} and writing down the edges encountered, with a minus sign if they are oriented against the direction of travel. More generally, we have the following.

Definition 1.17. Let G be an oriented graph with edges $e^1, \dots, e^r$. A l -chain on G is a formal sum

$$\lambda_1 e^1 + \lambda_2 e^2 + \dots + \lambda_r e^r$$

where each λ_i is an integer. When writing down 1-chains we omit any edges which have coefficient 0. The 1-chain in which all coefficients are zero is denoted 0. We define the *sum* of the 1-chains $\Sigma \lambda_i e^i$ and $\Sigma \lambda'_i e^i$ to be

$$\Sigma \lambda_i e^i + \Sigma \lambda'_i e^i = \Sigma (\lambda_i + \lambda'_i) e^i$$

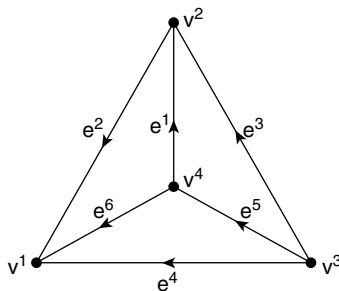
which is also a 1-chain. With this definition the set of 1-chains on G is an abelian group, denoted $C_1(G)$. (The verification that the 1-chains form an abelian group is left to the reader.)

In a similar way, a *0-chain* on G is a formal sum

$$\lambda_1 v^1 + \cdots + \lambda_m v^m$$

where $v^1, \dots, v^m$ are the vertices of G . 0-chains are added by adding coefficients, and the set of 0-chains is an abelian group, denoted $C_0(G)$.

Example 1.18. The 1-chain associated to a path is obtained in the manner described before 1.17. Note that the edges in a path need not be distinct (unless the path is simple), so that the final coefficient of an edge may be a number other than 0, 1 or -1 . For instance, the 1-chain associated with the path $v^4 e^1 v^2 e^2 v^1 e^6 v^4 e^1 v^2 e^3 v^3$ on the graph illustrated is $e^1 + e^2 - e^6 + e^1 - e^3 = 2e^1 + e^2 - e^3 - e^6$.



Definition 1.19. Let G be an oriented graph. For any edge $e = (vw)$ of G (regarded as a 1-chain), we define the *boundary* of e to be $\partial e = w - v$ (regarded as a 0-chain). The *boundary homomorphism*

$$\partial : C_1(G) \rightarrow C_0(G)$$

is defined by $\partial(\Sigma \lambda_i e^i) = \Sigma \lambda_i \partial(e^i)$. A 1-cycle on G is an element $c \in C_1(G)$ such that $\partial c = 0$. The group of 1-cycles on G is denoted $Z_1(G)$. Thus $Z_1(G)$ is the kernel of the homomorphism ∂ .

As an example, the 1-chain associated to any loop on G is always a 1-cycle; e.g. for the loop $v^1 e^2 v^2 e^1 v^4 e^5 v^3 e^4 v^1$ on the graph above, the associated 1-cycle is $-e^2 - e^1 - e^5 + e^4$ and

$$\begin{aligned}\partial(-e^2 - e^1 - e^5 + e^4) \\ = -(v^1 - v^2) - (v^2 - v^4) - (v^4 - v^3) + (v^1 - v^3) = 0.\end{aligned}$$

Thus cycles are generalizations of loops; the following theorem describes cycles completely on a connected graph.

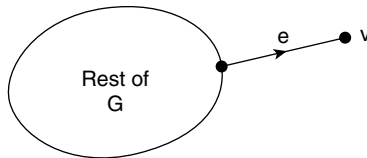
Now let G be a connected, oriented graph with cyclomatic number μ . Let z be a 1-cycle on G , and let $\ell^1, \dots, \ell^\mu$ be the loops described in 1.16, for some choice of maximal tree. We denote the 1-cycle associated to ℓ^i by z^i , the direction of travel round ℓ^i being chosen so that e^i occurs in z^i with coefficient $+1$.

Theorem 1.20. *There are unique integers $\lambda_1, \dots, \lambda_\mu$ such that*

$$z = \sum_{i=1}^{\mu} \lambda_i z^i$$

(if $\mu = 0$ this is to be read $z = 0$). In fact, λ_i is the coefficient of the edge e^i (see 1.16) in z .

Proof First suppose that G is a tree, i.e. that $\mu = 0$ and let z be a 1-cycle on G . We have to prove that $z = 0$, i.e. that there are no nontrivial 1-cycles on a tree. (This generalizes very minutely the fact that there are no loops on a tree.) The result is clear if G has no edges, so suppose it true for trees with $< k$ edges ($k \geq 1$) and let G have k edges. Let e be a non-inner edge of G , and v a free vertex of e (see 1.12(2)). The coefficient of e in z must be zero, since otherwise the coefficient of v in ∂z would not be zero. Hence z is a 1-cycle on the graph obtained by removing e and v from G . This graph is a tree with $k - 1$ edges; by induction $z = 0$.



Now suppose $\mu > 0$, and let z be a 1-cycle on G . Let λ_i be the coefficient of e^i in z , with e^i as in 1.16, and consider

$$z' = z - (\lambda_1 z^1 + \dots + \lambda_\mu z^\mu).$$

Certainly z' is a 1-cycle, since z and the z^i are. Also the coefficient of each e^i in z' is 0, by choice of the λ^i and since the coefficient of e^i in z^i is $+1$ while e^i does not occur in z^j for $j \neq i$. Thus z' is actually a 1-cycle on the graph obtained by removing the edges $e^1, \dots, e^\mu$ from G . This graph is a tree (maximal for G) and so by the first part of the proof $z' = 0$. This proves the required formula.

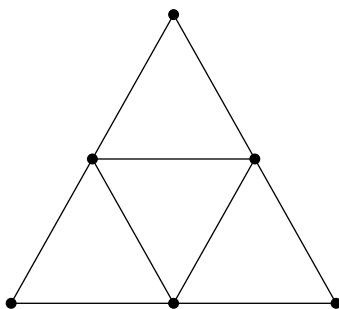
It remains to verify that the given expression for z is unique. To see this, suppose that

$$\sum_{i=1}^{\mu} \lambda_i z^i = \sum_{i=1}^{\mu} \lambda'_i z^i.$$

Comparing coefficients of e^i on the two sides gives $\lambda_i = \lambda'_i$. □

Remarks and Examples 1.21. (1X) Theorem 1.20 will be of considerable significance later when we come to calculate homology groups. For the present it shows that all 1-cycles, and hence all loops, on a connected graph, are in a precise sense combinations of the μ basic loops of 1.16. (It follows that all the Kirchhoff equations for loops are consequences of the equations for these basic loops.)

(2X) A *basis* for the 1-cycles on a connected graph G is a collection of 1-cycles of G in terms of which every 1-cycle on G can be expressed uniquely as a linear combination with integer coefficients. Thus $z^1, \dots, z^\mu$ in 1.20 is a basis. But not every basis arises in this way from a maximal tree. For instance, it is impossible to find edges e^1, e^2, e^3, e^4 of the graph shown, such that the resulting cycles z^1, z^2, z^3, z^4 are the basis given by the boundaries of the four small equilateral triangles.



(3) 1-cycles are defined only on an oriented graph. Suppose that G is an unoriented graph and G_1 and G_2 are oriented graphs each obtained from G by

orienting the edges of G arbitrarily. Then there is a natural isomorphism

$$C_1(G_1) \rightarrow C_1(G_2)$$

defined by $(vw) \rightarrow (vw)$ or (wv) according as the edge (vw) of G receives the same or opposite orientations in G_1 and G_2 . This clearly takes cycles to cycles, so that $Z_1(G_1) \cong Z_1(G_2)$. Orientation is a technical device introduced to enable the boundary homomorphism (1.19) to be defined, but the group of 1-cycles does not depend, up to isomorphism, on the orientation. In fact an equally satisfactory theory can be developed for unoriented graphs using 1-chains and 0-chains with coefficients in the group $\mathbb{Z}_2$ of integers modulo 2: this will be dealt with in a more general context in Chapter 8. The ' $\mathbb{Z}_2$ theory' is, however, inadequate for some situations we shall meet later, and that is why the other theory has been introduced from the start.

(4X) Suppose that G is a connected graph with cyclomatic number μ . Suppose that there exist edges $e^1, \dots, e^\mu$ and loops $\ell^1, \dots, \ell^\mu$ such that, for each i , e^i is an edge of ℓ^i but not of ℓ^j for $j \neq i$. Show that removing $e^1, \dots, e^\mu$ from G leaves a maximal tree.

(5X) A graph G is called *bipartite* if its vertex set can be partitioned into two disjoint subsets V_1, V_2 in such a way that every edge of G joins a vertex in V_1 to a vertex in V_2 . Show that G is bipartite if and only if every loop in G has an even number of edges. (Hence a tree is certainly bipartite!) [Hint for 'if'. Concentrate on one component G_1 of G . Define a homomorphism $\eta : C_1(G_1) \rightarrow \mathbb{Z}_2$ by $\eta c = 0$ or 1 according as the sum of coefficients in c is even or odd. Show (using 1.20) that $\eta z = 0$ for any 1-cycle z on G_1 . Now look at the 1-chains associated to paths from a fixed vertex v of G_1 to an arbitrary vertex w of G_1 . A direct proof can also be given by induction on $\mu(G_1)$.]

(6XX) Let $\ell^1, \dots, \ell^k$ be loops on an oriented connected graph and let $z^1, \dots, z^k$ be the 1-cycles given by these loops with a choice of direction. Let us call the loops 'independent' if, for all integers $\lambda_1, \dots, \lambda_k$, we have

$$\lambda_1 z^1 + \dots + \lambda_k z^k = 0 \Rightarrow \lambda_1 = \dots = \lambda_k = 0.$$

Find loops ℓ^1, ℓ^2, ℓ^3 on the complete graph with four vertices which are independent, but such that z^1, z^2, z^3 do not form a basis for the 1-cycles.

Show that two loops are independent if and only if they are distinct, and that three loops are independent if and only if none of the z^i is an integral linear combination of the other two.

(7X) Let G be an oriented graph with k vertices containing no oriented loop (an oriented loop is one in which all the edges point the same way round the

loop). Show that it is possible to label the vertices of G by $v^1, \dots, v^k$ in such a way that any oriented edge of G is of the form $(v^i v^j)$ with $i > j$. [Hint. Use induction on k . Note that, unless G has no edges, there must exist a vertex of G which is the beginning point of some edge but not the end point of any edge, for otherwise G would certainly contain an oriented loop.] The above result is used in Bernstein, Gel'fand and Ponomarev (1973) on p. 24 – note that they use ‘cycle’ in the sense of our word ‘loop’. This paper contains deep connexions between graph theory and linear algebra.

Having investigated the kernel of the boundary homomorphism (1.19), namely the 1-cycles, we shall say something about its image.

Definition 1.22. Let G be an oriented graph. The *augmentation map* is the homomorphism

$$\varepsilon : C_0(G) \rightarrow \mathbb{Z}$$

given by

$$\varepsilon(\sum \lambda_i v^i) = \sum \lambda_i.$$

Theorem 1.23. Let G be a connected oriented graph. Then the sequence

$$C_1(G) \xrightarrow{\partial} C_0(G) \xrightarrow{\varepsilon} \mathbb{Z}$$

is exact, that is to say $\text{Image } \partial = \text{Kernel } \varepsilon$.

Proof To show that $\text{Image } \partial \subset \text{Kernel } \varepsilon$ it is enough to verify that, for any edge e of G , $\varepsilon \partial e = 0$. This is immediate. For the converse, any element of $\text{Kernel } \varepsilon$ can be written in the form

$$\sum_{i=1}^{m-1} \lambda_i (v^i - v^m)$$

(where $v^1, \dots, v^m$ are the vertices of G), each λ_i being an integer. Since G is connected there is a path from v^m to v^i and the 1-chain c^i associated to this path has $\partial c^i = v^i - v^m$. Thus the above 0-chain equals

$$\partial(\sum \lambda_i c^i) \in \text{Image } \partial.$$

□

★ Planar graphs

A planar graph is a graph which is isomorphic to a graph in the plane $\mathbb{R}^2$. In this section we shall investigate some properties of planar graphs, mostly connected with the idea of cyclomatic number. At the end of the section we shall take up the question of whether our insistence that graphs have straight edges places any essential restriction on the class of planar graphs.

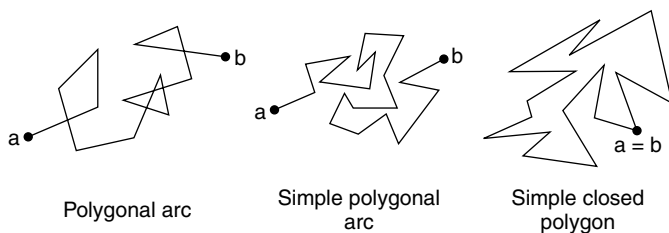
Some proofs will be omitted from this section and others will not be given in full detail; this is in order to avoid devoting undue space to what is, in fact, a subject of peripheral interest to the study of graphs in more general ‘surfaces’ in Chapter 9. At the end of the section the reader will need to be familiar with some point-set topology if he is to fill in all the details in the proof of 1.34. I hope, however, that the general idea of the proof will be clear to readers without such technical knowledge.

Definitions 1.24. A *polygonal arc* joining two points a and b in $\mathbb{R}^N$ is a sequence of straight segments placed end-to-end, starting at a and finishing at b . A polygonal arc is called *simple* if it has no points of self-intersection except that possibly $a = b$; thus a simple polygonal arc with $a \neq b$ can be thought of as a realization of the abstract graph given, for some $n \geq 2$, by

$$V = \{v^1, \dots, v^n\} \quad (n \text{ distinct vertices})$$

$$E = \{\{v^1, v^2\}, \{v^2, v^3\}, \dots, \{v^{n-1}, v^n\}\}.$$

A *simple closed polygon* is a simple polygonal arc with $a = b$; equivalently it is a realization of the abstract graph given, for some $n \geq 3$, by the same V and E as above, except that E has an additional edge $\{v^n, v^1\}$.



It is very plausible that a tree T realized in $\mathbb{R}^2$ will not separate the plane – i.e., that any two points of the plane not on T can be joined by a polygonal arc which does not intersect T . Indeed this can be proved quite easily by induction on the number of vertices of T , and the same proof works when T is replaced by a disjoint union of trees (a ‘forest’). It is also very plausible that a simple

closed polygon does separate the plane, and this is the content of the next theorem.

We call a subset of the plane *bounded* if it can be contained inside a suitable circle. Otherwise we call it *unbounded*. We say that a subset P of the plane *separates* the plane into two disjoint subsets I and O if the following hold.

- (i) $I \cup O = \mathbb{R}^2 \setminus P, I \cap O = \emptyset$.
- (ii) Any two points of I (resp. O) can be joined by a polygonal arc, hence by a simple polygonal arc, in I (resp. O).
- (iii) Any polygonal arc joining a point of I to a point of O intersects P .

(These say that $\mathbb{R}^2 \setminus P$ has precisely two ‘path components’.)

1.25. Jordan Curve Theorem for Polygons *Let P be a simple closed polygon in the plane. Then P separates the plane into two disjoint subsets I and O of which precisely one, O say, is unbounded. (We call I the ‘inside’ and O the ‘outside’ of P .) Furthermore if e is any edge of P and x and y are points of $\mathbb{R}^2 \setminus P$ sufficiently close to e , on opposite sides of it and away from the end-points, then precisely one of x, y belongs to I and (hence) the other belongs to O . $\square$*

A proof of 1.25 can be found in the book by Courant and Robbins listed in the References. The last sentence of 1.25 follows from their proof, or from the rest of the theorem together with the remarks on trees above.

A graph G in $\mathbb{R}^2$ will divide the plane into ‘regions’, two points not on G being in the same region if and only if they can be joined by a polygonal arc in $\mathbb{R}^2$ not intersecting G . Exactly one of these regions will be unbounded. (The graph G can be entirely contained inside a suitable circle, and any unbounded region will contain points outside this circle; on the other hand it is clear that any two points outside the circle belong to the same region.) Thus according to 1.25 a simple closed polygon will divide the plane into two regions. Our first objective is to count the regions for an arbitrary graph G .

Let e be an edge of a graph G in $\mathbb{R}^2$. Points near e and on one side of e will belong to one region, and e is called *adjacent* to that region; thus e is adjacent either to one region or to two different regions. It is clear that if e is part of a loop on G , then by 1.25 the regions on the two sides of e will be distinct; e will be adjacent to two regions. The converse is also true: if e is not part of any loop on G then e is adjacent to only one region. A proof is suggested in 1.30(3) below. Combining this with the fact, already noted in 1.9(4) for arbitrary graphs, that an edge of a graph belongs to a loop if and only if removing the edge does not increase the number of components of the graph, we have the following result.

Proposition 1.26. *Removing an edge e from a graph in $\mathbb{R}^2$ leaves the number of components of the graph unchanged if and only if it reduces the number of regions by one.*

It is now a straightforward matter to count the regions.

Theorem 1.27. *Let G be a graph in $\mathbb{R}^2$ with $\alpha_0(t)$ vertices, $\alpha_1(G)$ edges and $c(G)$ components. Then G divides the plane into $r(G) = \alpha_1(G) - \alpha_0(G) + c(G) + 1$ regions.*

Proof Let us remove the edges from G one at a time, leaving the vertices untouched, and note the effect on the value of $\alpha_1 - \alpha_0 + c + 1 - r$. In fact 1.26 says precisely that the value remains constant, and since when all the edges have gone it is $0 - \alpha_0(G) + \alpha_0(G) + 1 - 1 = 0$ it must have been 0 to begin with. This is the result. $\square$

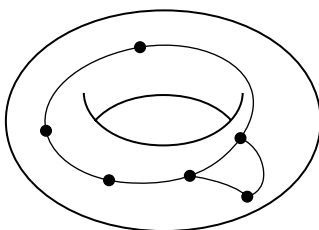
The formula for $r(G)$ given in 1.27 is known, at least for connected graphs ($c = 1$), as *Euler's formula*. For $c = 1$ it reads

$$\mathbf{1.28} \quad \alpha_0(G) - \alpha_1(G) + r(G) = 2,$$

or equivalently

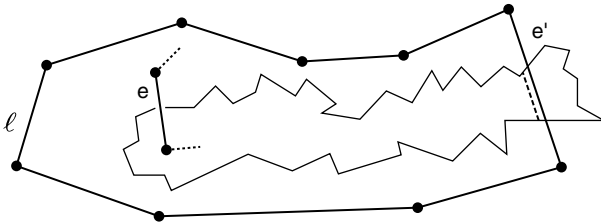
$$\mathbf{1.29} \quad r(G) = \mu(G) + 1.$$

Remarks and Examples 1.30. (1X) The most noticeable thing about the formula in 1.27 is that it exists at all: the number of regions depends only on the abstract structure of G and not on the way in which G is placed in the plane. It is possible to draw ‘graphs’ in surfaces other than the plane, for example in the torus surface. The edges will be curved lines on the surface, and again we can count regions. For the graph illustrated below, $\alpha_0 = 6$, $\alpha_1 = 7$, $c = 1$ and $r = 2$. However it is possible to find isomorphic graphs in the torus (these will have the same α_0 , α_1 and c), for which $r = 1$ or $r = 3$. Thus the number of regions does not depend only on the abstract structure of the graph. This phenomenon will be investigated in Chapter 9. What goes wrong when the argument leading to 1.26 is applied to graphs in the torus?

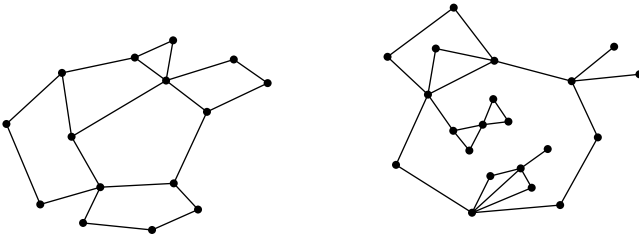


(2X) On the circumference of a circle in the plane n points are chosen and the chords joining every pair of points drawn. Assuming that no three of these chords are concurrent (except at one of the n points) show that the interior of the circle is divided into $1 + \binom{n}{2} + \binom{n}{4}$ regions. [Hint. Use Euler's formula. The number of points of intersection of chords inside the circle is $\binom{n}{4}$ since any four of the n points uniquely determine such a point of intersection. It is amusing to note that for $n = 1, 2, 3, 4, 5, 6$ the number of regions is 1, 2, 4, 8, 16, 31.]

(3X) Here is a suggestion for proving that, if an edge e of a graph G in $\mathbb{R}^2$ is not part of any loop on G , then the regions on the two sides of e are the same. First dispose of the case when G is a tree (e.g. using the remarks preceding 1.25), and then proceed by induction on the number of edges in G . Remove an edge e' from a loop ℓ on G , noting that $e \neq e'$ and e , apart possibly from its end-points, is entirely inside or entirely outside ℓ . Choose a polygonal arc from one side of e to the other, using the induction hypothesis. The only snag is that this arc may cross e' . It is not difficult to change the arc so that it doesn't cross e' ; a hint is given in the diagram.



We shall now look more closely at the formula 1.29 for connected graphs in $\mathbb{R}^2$. We already know, from 1.20, that there is a basis for the 1-cycles on G



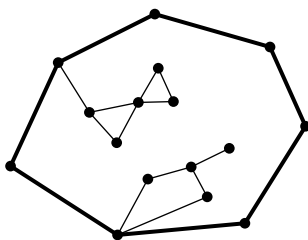
which consists of $\mu(G)$ elements. It is a consequence of a theorem of group theory (see A.30) that every basis has this number of elements. Now in the case of the left-hand illustration above (where $\mu(G) = 6$) it is very plausible that the boundaries of the six bounded regions give six cycles forming a basis for

the 1-cycles on G . (It is at least ‘clear’ that every loop on G can be obtained by combining these six loops suitably.) Perhaps this is not so obvious in the case of the right-hand illustration simply because it is not so clear which are the loops to take. If we can associate one basic loop with each bounded region, then this will confirm that $r(G) = \mu(G) + 1$, the ‘1’ on the right-hand side coming from the unbounded region. In fact such an association is possible, as stated in the next proposition, which will be given without proof.

Let G be a connected oriented graph in $\mathbb{R}^2$, and let R be one of the regions into which G divides $\mathbb{R}^2$. The *frontier* $\mathcal{F}(R)$ of R consists of those edges of G which are adjacent to R , and their end-points. Thus $\mathcal{F}(R)$ is a subgraph of G . Now suppose that R is a bounded region, and consider just the graph $\mathcal{F}(R)$ in $\mathbb{R}^2$, deleting the remainder of G .

Proposition 1.31. *The edges of $\mathcal{F}(R)$ which are adjacent to the unbounded region (when the remainder of G is deleted), together with their end-points, form a simple closed polygon. This gives us a loop on G ; the loops associated in this way to the various bounded regions give a basis for the 1-cycles on G .* $\square$

Note that the basis provided in this way may be different from any given by a maximal tree; see 1.21(2). As an example of 1.31, the frontier of one of the regions in the right-hand illustration above is drawn below, with the associated loop drawn heavily.



Another noteworthy point about the basis described in 1.31 is that any edge of G belongs to at most two of the basic loops. A remarkable theorem of S. Mac Lane (1937) asserts that the converse is true: if a connected graph G possesses loops $a^1, \dots, a^\mu$, whose corresponding 1-cycles form a basis for the 1-cycles on G and which have the property that each edge of G belongs to at most two of the loops a^i , then G is planar. Two other criteria for the planarity of a graph are known; one is due to Kuratowski and the other to Whitney (1932). See the books by Gross & Yellen, Diestel and Prasolov (*Combinatorial and Differential Topology*) listed in the References. There is also a proof of

Kuratowski's theorem which simultaneously proves Fáry's theorem (below) in Thomassen (1981).

We can make a further deduction from 1.31. Suppose that G (in $\mathbb{R}^2$) is connected, with $\mu(G) \geq 1$. Each loop associated to a bounded region will have at least three edges, so that the sum of the numbers of edges in these loops is at least 3μ . Also the frontier of the unbounded region will contain at least three edges, and adding these in gives a sum of at least $3\mu + 3$. This sum is $\leq 2\alpha_1(G)$ since an edge of G belongs to at most two of the loops and to at most one if the edge is adjacent to the unbounded region. From this we deduce the following.

Proposition 1.32. *If G is a connected graph in $\mathbb{R}^2$ satisfying $\mu(G) \geq 1$, then $3\alpha_0(G) - \alpha_1(G) \geq 6$. $\square$*

The case $\mu(G) = 0$ is left to the reader. As an example of the proposition let G be the complete graph with five vertices. For this G , $\alpha_0 = 5$, $\alpha_1 = 10$. (Also $\mu = 6 \geq 1$.) Thus $3\alpha_0 - \alpha_1 < 6$, and G is not planar.

Example 1.33. (X) Show that if G is a connected bipartite graph (1.21(5)) in $\mathbb{R}^2$ satisfying $\mu(G) \geq 1$, then $2\alpha_0(G) - \alpha_1(G) \geq 4$. Find a bipartite graph which is not planar.

Given a graph G we can 'subdivide' it by adding extra vertices at interior points of edges; this automatically increases the number of edges by the same amount, thus leaving $\mu(G)$ unchanged. For example if we subdivide the complete graph with five vertices by adding a single vertex on one of the edges the new graph has $\alpha_0 = 6$, $\alpha_1 = 11$ and it is no longer true that $3\alpha_0 - \alpha_1 < 6$. Thus 1.32 does not show that the subdivided graph is non-planar. This indicates, of course, how feeble the result 1.32 is; all the same it is a priori conceivable that by adding enough extra vertices on the edges of a non-planar graph we could make it planar – this is rather like trying to draw it in the plane with curved edges, since these could be approximated by broken lines.

A theorem of K. Wagner (1936) and I. Fáry (1948), known as Fáry's Theorem, implies that this does not happen; if some subdivision of a graph G is planar, then G is planar. It would take us too far out of our way to give a full and rigorous proof of Fáry's theorem, since this would involve the formal introduction of some notions of point-set topology with which we are not otherwise concerned. However there follows a proof that is complete except for these details. It is a variant of Fáry's original proof, and was found by P. Damphousse. In fact Damphousse's argument proves a stronger result than that proved by Fáry (this stronger result is stated in Fáry's paper). (See also the books of Prasolov (*Combinatorial and Differential Topology*) and Gross & Yellen listed in the References; also Bryant (1989).)

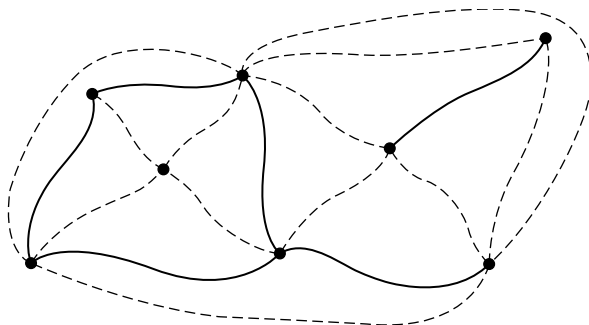
We shall consider ‘curved graphs’ in the plane which differ from the graphs considered so far only in that their edges are allowed to be curved lines (precisely: homeomorphic images of the closed interval $[0, 1]$) instead of straight segments. Thus a *curved graph* $\mathcal{G}$ consists of finite set of points (vertices) in the plane, some pairs of distinct vertices being joined by curved lines (edges) which do not intersect except possibly at end-points. An edge does not intersect itself either: it is a simple curve. Furthermore any particular pair of vertices is joined by at most one edge. It is possible to formulate and prove Euler’s formula 1.27 for curved graphs: both formulation and proof require the most general form of the Jordan Curve Theorem which states, roughly speaking, that a simple closed curve always separates the plane into two regions. (See the books of Prasolov (*Combinatorial and Differential Topology*) and Wall listed in the References.) In this sketch we shall assume the validity of Euler’s formula for curved graphs, and of some other ‘plausible’ results to be mentioned later.

The weak form of Fáry’s theorem asserts that the abstract graph underlying $\mathcal{G}$ can be realized in $\mathbb{R}^2$ (see 1.3), i.e. that there is a (straight) graph G in $\mathbb{R}^2$ with the same abstract structure as $\mathcal{G}$. The strong form is as follows.

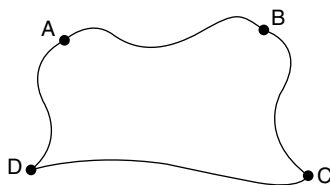
1.34. Fáry’s Theorem on straightening planar graphs *With the above notation, there is a homeomorphism of the plane (that is, a bijective continuous map $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ with continuous inverse) mapping $\mathcal{G}$ onto a (straight) graph G in such a way that vertices of $\mathcal{G}$ are taken to vertices of G and edges to edges.*

Proof The proof is in several parts. First we make changes to $\mathcal{G}$ to simplify its structure (I); (II)–(IV) are consequence of these changes. (V) is the central argument (by induction on the number of vertices of $\mathcal{G}$) and (VI) is a small result needed in (V). We assume throughout that $\mathcal{G}$ has at least four vertices, the other cases being easily disposed of.

(I) We turn $\mathcal{G}$ into a *triangular* graph $\mathcal{G}^\Delta$, that is a curved graph for which each one of the regions into which $\mathcal{G}^\Delta$ divides the plane has frontier (see before 1.31) a simple closed curve consisting of precisely three edges of $\mathcal{G}^\Delta$. This is achieved by a sequence of operations on $\mathcal{G}$, as follows. Select any two vertices of $\mathcal{G}$ which are on the frontier of the same region and which are not already connected by any edge of $\mathcal{G}$; connect them by a simple curve within the region to give a graph $\mathcal{G}'$. Now select a pair of vertices of $\mathcal{G}'$ which are on the frontier of the same region defined by $\mathcal{G}'$ and are not connected by any edge of $\mathcal{G}'$ and connect them within the region. This procedure is continued until it is no longer possible to find a pair of vertices of the sort described – this must eventually happen since no new vertices are introduced and consequently the number of possible edges is bounded. An example is drawn below; the dotted edges are the ones introduced.



To see that the result is triangular, suppose that a region has frontier consisting of say four edges, as in the diagram below. By the construction A and C must be joined by an edge outside the region and so must B and D . But this is impossible in $\mathbb{R}^2$. (This is ‘obvious’ but formally requires the Jordan Curve Theorem.)



(II) In any triangular graph with at least four vertices there can be no vertex of order 0, 1 or 2. To see, for example, that there can be no vertex of order 2 suppose that there is one – B say, in the diagram above. As there is at least one vertex other than A, B, C there must be such a vertex on the frontier of one of the regions adjacent to the arc ABC . But this vertex is not joined to B , so (I) has not been completed.

(III) In any triangular graph with at least four vertices there are at least four vertices of order ≤ 5 . From the triangularity it follows that $\mathcal{G}$ is connected and $3r = 2\alpha_1$ (r being the number of regions) and this together with Euler’s formula 1.28 shows $6\alpha_0 - 2\alpha_1 = 12$. Now writing N_i for the number of vertices of order i (so that $N_0 = N_1 = N_2 = 0$ from (II)), we have

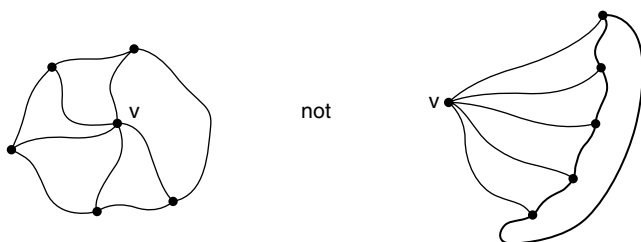
$$\alpha_0 = N_3 + N_4 + N_5 + \cdots \quad \text{and} \quad 2\alpha_1 = 3N_3 + 4N_4 + 5N_5 + \cdots$$

so that $12 = 6\alpha_0 - 2\alpha_1 = 3N_3 + 2N_4 + N_5 - N_7 - 2N_8 - \cdots \leq 3N_3 + 2N_4 + N_5$.

Hence $N_3 + N_4 + N_5 \geq \frac{1}{3}(3N_3 + 2N_4 + N_5) \geq 4$, as required.

(IV) There is at least one vertex v of $\mathcal{G}^\Delta$, of order ≤ 5 and not on the frontier of the unbounded region. This is because there are precisely three vertices of

$\mathcal{G}^\Delta$ on the frontier of the unbounded region, and by (III) at least four vertices of $\mathcal{G}^\Delta$ of order ≤ 5 . Now consider the edges with v as one end-point; they can be ordered, say anticlockwise, round v . (This statement is not as innocent as it looks, for the edges could for example spiral round v in a complicated manner. What they cannot do is cross each other, and that is what makes the result, which follows from the Jordan Curve Theorem, true.) Consecutive pairs of these edges will terminate in pairs of vertices, and these pairs will be joined by edges of $\mathcal{G}^\Delta$ which will form a simple closed curve $\mathcal{C}$ enclosing v .



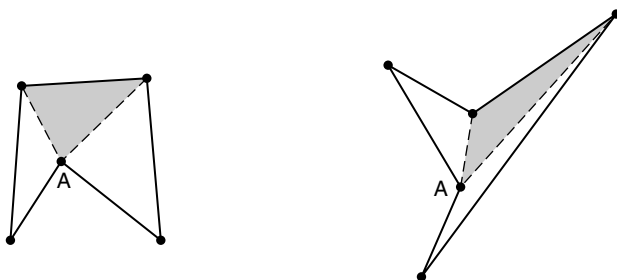
(It is not hard to see that the right-hand picture can only arise when v is on the frontier of the unbounded region.)

(V) This is the induction argument by which the theorem is proved. Suppose 1.34 is true whenever $\mathcal{G}$ has $< k$ vertices ($k \geq 4$) and let $\mathcal{G}$ have k vertices. (I)–(IV) apply to $\mathcal{G}$, which turns into a triangular graph $\mathcal{G}^\Delta$; we then remove the v of (IV) together with its attendant edges, leaving a graph $\mathcal{G}_1$, say. Now use the induction hypothesis to choose a homeomorphism h of the plane to itself mapping $\mathcal{G}_1$ on to a straight graph G_1 . This will take $\mathcal{C}$ (see (IV)) to a simple closed polygon C say, with 3, 4 or 5 sides and with no vertex inside it since h will preserve the interior of a simple closed curve. It remains to ‘fill in’ the missing vertex and edges of $\mathcal{G}$ and change h inside $\mathcal{C}$ so that it maps them to a point P inside C and straight segments joining P to the vertices around C . Thus we need, in the first place, the existence of point P within C from which all vertices of C are ‘visible’. This is dealt with in (VI).

Having found such a P it is possible to redefine h inside $\mathcal{C}$ as above. (First we ensure that $h(v) = P$, then that the curved edges through v are mapped to the straight edges through P . It then remains to extend h to the interiors of the ‘curved triangles’ inside $\mathcal{C}$. This is a standard procedure in point-set topology and we shall not go into the details.) Since $\mathcal{G}$ is a subgraph of $\mathcal{G}^\Delta$, this completes the induction, for h will automatically take $\mathcal{G}$ on to a straight graph.

(VI) Within any simple closed polygon C with 3, 4 or 5 sides there is always a point P from which all the vertices of C are visible. The proof of this is left as an exercise, with the following hint for a pentagon. Divide the interior of C

into three triangles by two straight segments connecting vertices of C and lying entirely inside C . (It can be proved, in fact, that the interior of any simple closed polygon with n sides can be divided into $n - 2$ triangles by drawing $n - 3$ such segments, intersecting only at vertices of the polygon.) Since there are three triangles one of them must be 'central' in the sense that it is joined to each of the other two along an edge. In the diagrams the central triangle is shaded.



It is not difficult to find points within the central triangle close to the vertex A from which all vertices of the pentagon are visible.

Note that pentagons are a critical case: there exist hexagons within which there is no point from which all six vertices are visible.

This completes the sketch of a proof of Fáry's Theorem. ★

□

★ Appendix on Kirchhoff's equations

In this appendix we shall present a proof of the fact that Kirchhoff's equations (see the section starting after 1.7) determine the currents uniquely for any circuit. Consider the basic loops $\ell^1, \dots, \ell^\mu$ described in 1.16. The $(K1)$ equation of the loop ℓ^i will be written $L_i = b_i$ where L_i is the part involving the (unknown) currents and b_i is the total e.m.f. in the loop. As has already been noted in 1.21(1) the $(K1)$ equation of any loop is a consequence of these μ equations; it was also noted during the earlier discussion that all but one of the $(K2)$ equations are independent. Writing $V_i = 0$ for the equation of the vertex v^i we have the following, which establishes the required result.

Proposition 1.35. *The $\alpha = \alpha_1$ equations*

$$L_1 = b_1, \dots, L_\mu = b_\mu, \quad V_2 = 0, \dots, V_n = 0 \quad (n = \alpha_0)$$

are independent.

Proof Suppose that

$$\lambda_1 L_1 + \cdots + \lambda_\mu L_\mu = v_2 V_2 + \cdots + v_n V_n. \quad (*)$$

We prove that $\lambda_1 = \cdots = \lambda_\mu = 0$ which, in view of the independence of $V_2, \dots, V_n$, implies the result.

We introduce some new notation at this point. It is not strictly necessary to the argument, but allows a more succinct presentation. The ideas of ‘cochain’ and ‘coboundary’ are of great importance in algebraic topology, but we shall not have occasion to make use of them apart from in this one case.

In order to allow the resistances in the wires to be real numbers and not just integers, it is convenient to work with ‘real chains’, that is linear combinations of vertices or edges in which the coefficients are real numbers. In 1.17 the λ_i become elements of $\mathbb{R}$, the real numbers, and $C_0(G)$ and $C_1(G)$ become real vector spaces. Likewise in 1.19 the boundary homomorphism ∂ becomes a linear map. We shall keep to the old notation, but, just for this proof, work over $\mathbb{R}$ instead of $\mathbb{Z}$.

A *0-cochain* is a linear map $\gamma^0 : C_0(G) \rightarrow \mathbb{R}$ and a *1-cochain* is a linear map $\gamma^1 : C_1(G) \rightarrow \mathbb{R}$. The value of a cochain γ on a chain c is denoted $\langle \gamma, c \rangle \in \mathbb{R}$, and the i -cochains form a real vector space $C^i(G)$ ($i = 0, 1$) with addition and scalar multiplication defined by

$$\langle \gamma + \gamma', c \rangle = \langle \gamma, c \rangle + \langle \gamma', c \rangle; \quad \langle \lambda \gamma, c \rangle = \lambda \langle \gamma, c \rangle.$$

There is a *coboundary homomorphism*

$$\delta : C^0(G) \rightarrow C^1(G)$$

defined by

$$\langle \delta \gamma^0, c_1 \rangle = \langle \gamma^0, \partial c_1 \rangle.$$

Now writing x_j for the (unknown) current in the oriented edge e^j and r_j for the resistance of this (unoriented) edge, L_i will have the form

$$L_i = \sum_{j=1}^{\alpha} \epsilon_{ij} r_j x_j \quad (\epsilon_{ij} = -1, 0 \text{ or } 1).$$

Taking the direction round the loop ℓ^i to be given by the orientation of the edge e^i , it follows that $\epsilon_{ij} = 0$ if $i \neq j$ and $j \leq \mu$, while $\epsilon_{ii} = +1$. This is because e^j

does not occur in any of the loops ℓ^i for $i \neq j \leq \mu$. Hence the $\mu \times \alpha$ matrix $A = (\epsilon_{ij})$ has the $\mu \times \mu$ identity matrix as its first μ columns. This will be of significance later.

With L_i we associate the 1-cochain, also called L_i , which takes the edge e^j to $\epsilon_{ij}r_j \in \mathbb{R}$. In the same way we can associate a 1-cochain with V_i , and it is straightforward to check that $V_i = \delta v^i$, where v^i is the 0-cochain taking the value 1 on the vertex v^i and 0 on every other vertex. Thus the equation (*) above says precisely that $\lambda_1 L_1 + \cdots + \lambda_\mu L_\mu$ is in the image of δ .

It follows from the formula for δ that if we evaluate $\lambda_1 L_1 + \cdots + \lambda_\mu L_\mu$ on any cycle the result will be zero. Let us evaluate it on the cycles $z^1, \dots, z^\mu$ given by the basic loops $\ell^1, \dots, \ell^\mu$. The result is μ linear equations in $\lambda_1, \dots, \lambda_\mu$ and a little calculation reveals that the coefficient matrix is $(\langle L_i, z^j \rangle)$. With A as above and $R = \text{diag}(r_1, \dots, r_\alpha)$ this coefficient matrix is precisely $ARA^\top$ ($\top$ for transpose). We shall show that this matrix is nonsingular.

Now each resistance is strictly positive, so we can define a nonsingular matrix $S = \text{diag}(\sqrt{r_1}, \dots, \sqrt{r_\alpha})$. Clearly $ARA^\top = (AS)(AS)^\top$. However A has rank μ since it contains the $\mu \times \mu$ identity matrix. Hence AS has rank μ . It is now a standard fact of linear algebra that, AS being real, the product with its transpose is nonsingular. (One way to prove this is to note that $(AS)(AS)^\top$, being symmetric, is congruent to a diagonal matrix. This is a standard result of linear algebra.) Thus $\lambda_1 = \cdots = \lambda_\mu = 0$, as required. $\square$

It is not difficult to work this argument backwards and show that, if $ARA^\top$ is singular, then the Kirchhoff equations are dependent or contradictory. (The backward argument uses the fact that a cochain which vanishes on every cycle is necessarily a coboundary. This holds over $\mathbb{R}$, but not over $\mathbb{Z}$.) Thus the nonsingularity of $ARA^\top$ is necessary and sufficient for the Kirchhoff equations to determine the currents uniquely. It is a curious fact that this condition is much weaker than the condition ' $r_i > 0$ for all i ' which was used at the end of the proof. For example when $\mu = 1$ it says that the sum of the resistances in the unique loop should be non-zero.

In Kirchhoff's original 1847 paper, he shows how to determine the currents (in the physical situation $r_i > 0$) and also shows that all the $(K-1)$ equations depend on the μ equations for the basic loops. Parts of this paper are readily available in translation in the book of Biggs, Lloyd & Wilson listed in the References. Other useful references are the books of Bollobás, Wilson & Beinecke and Tutte. ★